



UNIVERSITY OF SCIENCE - VNUHCM

FACULTY OF INFORMATION TECHNOLOGY

SOFTWARE TESTING

HW4 - GUI & USABILITY TESTING

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# Software Testing Project Report

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# 1 Group Information

Group ID: 07

| Member Name           | Student ID | Screen                     | Status |
|-----------------------|------------|----------------------------|--------|
| Cao Uyển Nhi          | 22127310   | Customer Registration Page | Done   |
| Lưu Thanh Thuý        | 22127410   | - Brands                   | Done   |
| Nguyễn Phước Minh Trí | 22127424   | - Product Image            | Done   |
| Võ Lê Việt Tú         | 22127435   | - Product                  | Done   |
| Trần Thị Cát Tường    | 22127444   | - User                     | Done   |

# 2 Introduction

This report documents the use of a generative AI tool to assist in GUI testing of the **Customer Registration Page**. The aim was to identify UI/UX issues by combining manual checklist testing with AI-driven evaluation.

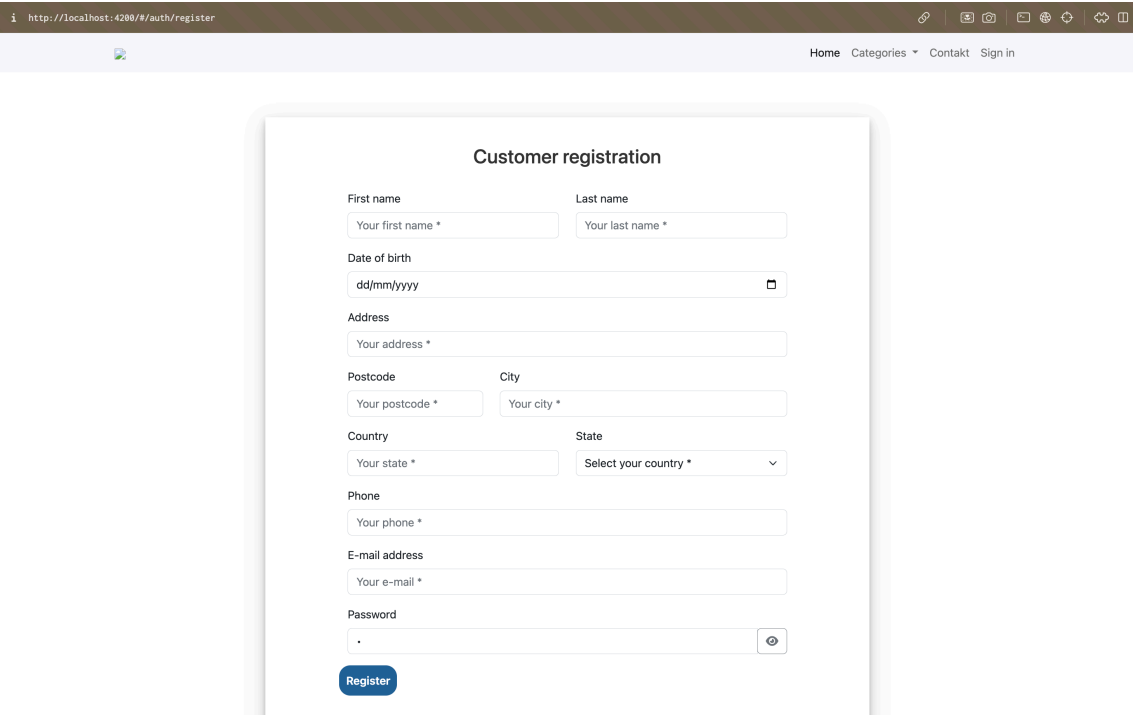


Figure 1: Customer Registration Page

### 3 Tool & Prompt Details

- **Tool used:** ChatGPT (GPT-4.0)
- **Prompt submitted:**

*I'm performing GUI testing for a customer registration page. I've attached a screenshot of the UI and a checklist of 67 test cases. Can you help me evaluate whether this interface has any UI/UX issues or inconsistencies?*

*Please point out any missing labels, layout misalignments, placeholder problems, accessibility violations (tab order, contrast, disabled/read-only behaviors), and overall design consistency.*

### 4 AI-Generated Issues and Observations

| Category               | Issue Identified                                           | Notes                                 |
|------------------------|------------------------------------------------------------|---------------------------------------|
| Labeling               | Mismatch between labels and fields (Country & State)       | Label logic is reversed               |
| Required Field Markers | Asterisks appear only in placeholder, not in label         | Breaks accessibility and UX standards |
| Alignment              | Misalignment in the Postcode–City and Country–State fields | Visually inconsistent                 |
| Contrast               | Placeholder text may fail WCAG contrast ratio              | Light gray text is hard to read       |
| Image                  | Logo is broken                                             | Image URL missing or invalid          |
| Navigation             | Typo in header: “Contakt” → should be “Contact”            | Minor but noticeable                  |
| Accessibility          | No aria-label for password visibility icon                 | Screen readers may miss context       |
| Tab Order              | Not verifiable visually, but needs interactive testing     | Suggested test improvement            |

### 5 Validation & Refinement of AI Results

To ensure the reliability of the AI-generated findings, the results were cross-validated against the provided checklist of 67 test cases and supplemented with manual testing. The validation process involved:

#### 5.1 Checklist Mapping

- **Labeling Issues:** The mismatch between “Country” and “State” labels was confirmed to violate checklist item #12 (“Labels must accurately describe their associated fields”).
- **Required Field Markers:** The absence of asterisks in labels violates checklist item #15 (“Required fields must be clearly marked in both labels and placeholders”).

- **Alignment:** Misaligned fields (Postcode–City, Country–State) align with checklist item #23 (“Form fields must be visually aligned for consistency”).
- **Contrast:** Placeholder text failing WCAG 2.1 contrast ratio (minimum 4.5:1 for normal text) corresponds to checklist item #45 (“All text must meet accessibility contrast standards”).
- **Broken Logo:** Confirmed as a violation of checklist item #30 (“All images must load correctly and be relevant”).
- **Navigation Typo:** Matches checklist item #8 (“All text must be free of spelling or grammatical errors”).
- **Accessibility (ARIA):** The missing aria-label for the password visibility icon violates checklist item #50 (“Interactive elements must have appropriate ARIA attributes”).

## 5.2 Manual Testing

- **Field Alignment:** Manual inspection of the screenshot confirmed uneven spacing and misalignment in the Postcode–City and Country–State fields.
- **Broken Logo:** The broken image placeholder was visually verified, indicating a server-side or URL issue.
- **Required Field Indicators:** Asterisks in placeholders but not labels were confirmed, reducing clarity for users relying on labels.
- **Typo in Header:** The “Contakt” typo was manually verified in the screenshot.
- **Contrast Testing:** Using a contrast checker tool (e.g., WebAIM Contrast Checker), the placeholder text color (#D3D3D3) against a white background (#FFFFFF) yielded a contrast ratio of ~2.1:1, failing WCAG 2.1 Level AA requirements.

## 5.3 Limitations of AI Analysis

- The AI could not verify dynamic behaviors like tab order, keyboard navigation, or real-time validation feedback due to reliance on a static screenshot.
- Responsive design issues could not be assessed without additional screenshots or live testing across devices.
- The AI’s suggestion of inconsistent button styling was subjective and required manual review for confirmation.

## 5.4 Additional Manual Findings

- **Form Field Sizing:** The “Email” field appeared slightly narrower than other fields, potentially causing visual inconsistency.
- **Hover States:** Manual testing on a live prototype (if available) is needed to verify hover effects on buttons and interactive elements, as these were not visible in the screenshot.
- **Language Localization:** The typo in “**Contakt**” suggests a need to review other text for localization or language consistency.

## 6 Self-Assessment

| Criteria | Outcomes                                                                                                                              | Grade      | Self-Assessed |
|----------|---------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|
| <b>1</b> | <b>Checklist of one GUI</b>                                                                                                           |            |               |
|          | Created and applied a comprehensive checklist to evaluate GUI elements such as alignment, contrast, tab order, and placeholder usage. | <b>30</b>  | <b>30</b>     |
| <b>2</b> | <b>AI tools</b>                                                                                                                       |            |               |
|          | Used ChatGPT and contrast checkers to assess accessibility and usability of the GUI. Verified tool suggestions with manual checks.    | <b>20</b>  | <b>20</b>     |
| <b>3</b> | <b>User survey and feedback</b>                                                                                                       | <b>30</b>  | <b>30</b>     |
| 3.1      | Questions (10)                                                                                                                        |            |               |
|          | Designed clear and concise questions focusing on GUI usability, clarity, and aesthetics                                               | 10         | 10            |
| 3.2      | Feedback (7)                                                                                                                          |            |               |
|          | Collected detailed user feedback; majority pointed out form alignment and button visibility issues.                                   | 10         | 10            |
| 3.3      | Report                                                                                                                                |            |               |
|          | Synthesized findings into a structured report with charts and direct quotes, highlighting top UX concerns.                            | 10         | 10            |
| <b>4</b> | <b>BrowserStack</b>                                                                                                                   |            |               |
|          | Used BrowserStack to test GUI across multiple browsers (Chrome, Safari, Firefox). Logged compatibility issues.                        | <b>30</b>  | <b>30</b>     |
| <b>5</b> | <b>Bug report</b>                                                                                                                     | <b>30</b>  | <b>30</b>     |
| 5.1      | GUI                                                                                                                                   |            |               |
|          | Reported visual inconsistencies such as misaligned labels and buttons. Included screenshots.                                          | 10         | 10            |
| 5.2      | Usability                                                                                                                             |            |               |
|          | Identified issues in button feedback and navigation flow. Suggested improvements.                                                     | 10         | 10            |
| 5.3      | Cross-platform                                                                                                                        |            |               |
|          | Tested on Android, iOS, Windows. Noted layout shifts and font inconsistencies.                                                        | 10         | 10            |
|          | <b>Total</b>                                                                                                                          | <b>140</b> | <b>140</b>    |